

# Quiz 11

**STAT 109: Introductory Biostatistics**

# Quiz 11

## Quiz 11 practice problems

These practice problems cover **Lecture 28** and **Lecture 29**.

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### Instructions

- Work **without the solutions** first, then check your work in Canvas.
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### Part A: Study design and causal questions (Lecture 28)

#### A1.

Researchers randomly assign volunteers with mild seasonal allergies to use either a new nasal spray (**Spray**) or an identical-looking placebo (**Control**) for four weeks. They record whether each person reports “clear improvement” (yes/no). Which description best fits this study?

- a) Randomized controlled experiment (RCT)
  - b) Observational study
  - c) Simple Random Sampling
  - d) Parameter
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#### A2.

Suppose the spray group has a higher rate of “clear improvement” than the control group, and a statistical test yields a small  $p$ -value. In **one sentence**, what explanation does the  $p$ -value address? (Choose the best option.)

- a) The probability that the spray **causes** improvement in any one person.
  - b) How compatible the data are with a “**by chance alone**” explanation for the observed difference between groups, under the test’s assumptions and null hypothesis.
  - c) Proof that no confounding occurred.
  - d) The probability that the null hypothesis is true.
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### A3.

A newspaper headline reads: “People who drink more green tea live longer, study finds.” The underlying study measured tea intake and survival in a large **observational** study (no random assignment of tea drinking). Which conclusion is **most appropriate** for the researchers to emphasize in the discussion?

- a) Green tea **caused** longer life in the study population.
  - b) Higher tea intake is linked with longer survival; a **causal** interpretation requires ruling out confounding and other biases, which this design does not do by itself.
  - c) Because the sample was large, **causation** is established.
  - d) It’s not ethical to force people to drink green tea so this study provides high quality evidence by allowing people to chose to drink green tea or not.
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### A4.

Which design most directly supports causal language about the effect of a treatment on an outcome (for people like those enrolled)?

- a) Volunteers choose treatment A vs B; outcomes are compared.
  - b) A randomized controlled trial.
  - c) A survey asking people whether they used treatment A and whether they feel better.
  - d) A chart review comparing outcomes at two hospitals that use different protocols, with no randomization.
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### A5.

Describe the main difference between an observational study and a randomized controlled trial (RCT) when comparing an explanatory variable (treatment or exposure) to a response variable, focusing on how participants end up in the comparison groups.

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### A6.

A statistical analysis finds an association between an explanatory variable  $X$  and a response  $Y$ .

- (a) If the study is observational, list at least three kinds of explanations (use short labels or phrases) for why  $X$  and  $Y$  might appear to be linked
  - (b) If the study is a well-conducted RCT comparing assigned treatment to control and a difference in  $Y$  is observed, what two broad explanations are possible?
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## Part B: Confounding — definitions and recognition (Lecture 29)

### B1.

A **confounder** is best described as a variable that....

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## B2.

Researchers compare mean blood pressure between two clinics. Clinic A has lower mean blood pressure than Clinic B. They ignore age: Clinic A serves many young adults; Clinic B serves many older adults, and age is strongly related to blood pressure. Age here acts as:

- a) The primary outcome of interest.
  - b) A confounder of the clinic–blood pressure association.
  - c) A mediator caused by clinic that explains blood pressure.
  - d) Random chance only.
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## B3.

Why does random assignment of treatment in an RCT reduce the risk of confounding for the treatment–outcome comparison (as taught in Lecture 28), compared with letting people choose their treatment?

- a) Randomization guarantees there are zero confounders.
  - b) Randomization tends to balance both measured and unmeasured baseline factors across arms on average, so treatment groups are more comparable at the start.
  - c) Randomization removes the need to record the outcome.
  - d) Randomization replaces the need for any statistical test.
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## B4.

Two variables are associated in the data. You are told the explanatory variable **causes** a third variable, and that this third variable **causes** the response variable, and the explanatory variable does **not** directly cause the response except **through** that third variable. That third variable is:

- a) A confounder.
  - b) Not a confounder.
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## B5.

Observational study. Researchers record whether adults choose to take a daily vitamin D supplement ( $X$ ) and measure femoral neck bone density as a numerical response  $Y$  (same clinic visit as supplement status).

Which of the following variables could be a confounder of the association between  $X$  and  $Y$ ? Select all that apply.

- a) Age (related to both supplement habits and bone density)
- b) Femoral neck bone density  $Y$  itself
- c) Overall diet quality
- d) Sex assigned at birth

- e) Parathyroid hormone (PTH) measured at the same visit, on the biological pathway by which vitamin D status influences bone turnover
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### B6.

Observational study.  $X$  = whether a child's home is within 500 m of a busy road;  $Y$  = whether the child has ever been given a diagnosis of asthma by a clinician.

Which of the following could be a confounder of an observed link between  $X$  and  $Y$ ? Select all that apply.

- a) Household income (related to which neighborhoods families can afford and to healthcare access)
  - b) The child's asthma diagnosis  $Y$
  - c) Parental cigarette smoking in the home
  - d) Traffic-related air pollution concentration estimated at the home (often "near road  $\rightarrow$  pollution  $\rightarrow$  asthma")
  - e) Child's age in years
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### B7.

You are analyzing an observational study with an explanatory variable  $X$  and a response  $Y$ . You identify a variable  $C$  that could be a confounder, and you can measure  $C$  for everyone in the dataset. Which option best describes a central idea of subclassification?

- a) Drop all rows with  $C$  missing and rerun only a  $p$ -value on the full sample.
  - b) Split the data by the categories of  $C$  and compare the  $X$ - $Y$  relationship within each level of  $C$  (same value of the confounder), e.g. tables or plots or models separately in each category of  $C$ .
  - c) Replace  $X$  with  $C$  so the analysis no longer mentions the explanatory variable.
  - d) Collapse  $Y$  and  $C$  into a single composite score before making any plots or computing a  $p$ -value.
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## Part C: Critique claims in context (Lectures 28–29)

### C1.

A campus survey finds that students who report more hours of sleep also report higher GPAs. A mattress company blog claims: "Sleep more and your grades will improve." Write two or three sentences explaining why this claim may be too strong, naming at least one alternative explanation besides "sleep causes GPA".

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### C2.

An observational study finds that people who enroll in a free fitness program have better cardiovascular health one year later than people who do not enroll. The program is not randomized. Give one reason why the conclusion "the program caused better health" might be unwarranted without further evidence.

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